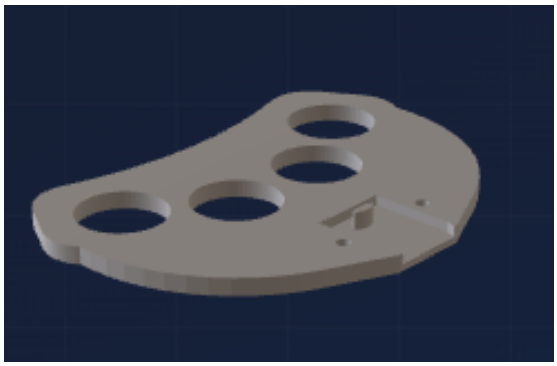
#1 - Extended keys

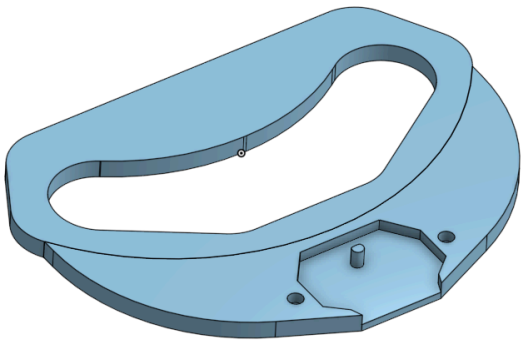
Penn Robotics was asked to design and print an extension for everyday keys to help people who may not normally be able to use a key. The team went through multiple iterations before deciding on a final design.

Original designs-

 <- This design was changed because the bump where the hand went was uncomfortable to grasp

 <- The knuckles were very comfortable to wear, but aren't for all hand sizes

Attempt #1(above)- the area for the key wasn't quite wide enough for a normal key. The column in the center was a little too wide to fit most keys.

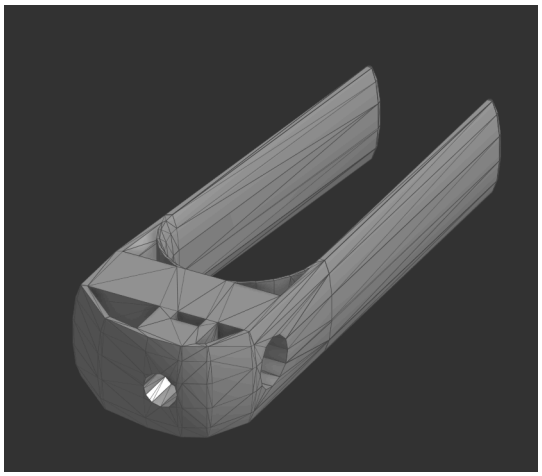


Attempt #2(above)- Changes: Size of key area, size of keyhole for positioning, angle of grip.

#2 - Palm Pen Holder

Penn Robotics was asked to redesign the Palm Pen Holder from https://www.makersmakingchange.com/. The redesign was for a young adult with cerebral palsy who had difficulties writing. We wanted to make the grip much more comfortable and much bigger so it fit in the hand better.

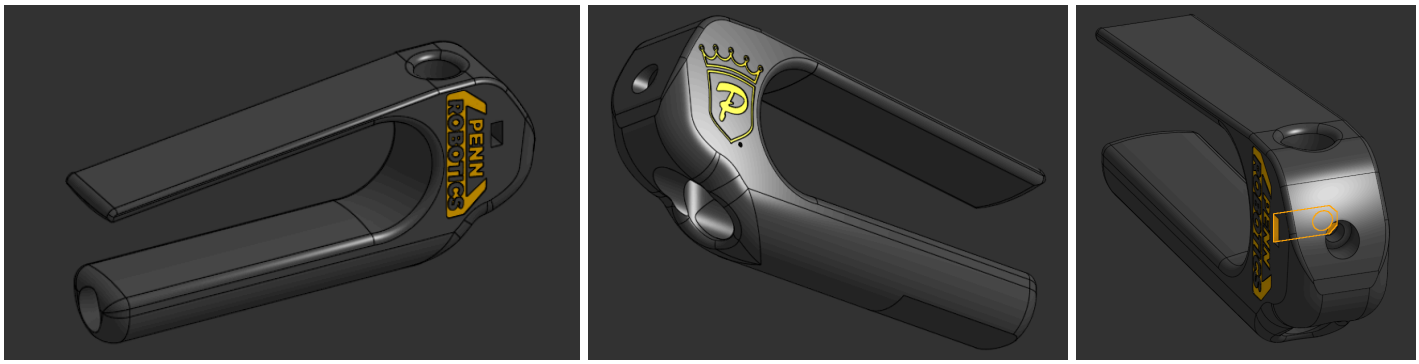
Original design:



Source: https://www.makersmakingchange.com/product/palm-pen-holder/01tJR00000068z8YAA

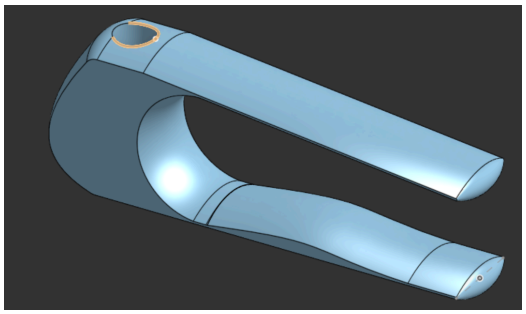
Issues: This design was not ergonomic for the hand; it had little surface area for the palm and fingers. It did not conform to the natural gripping shape of the hand, leading to cramping and worse handwriting over time.

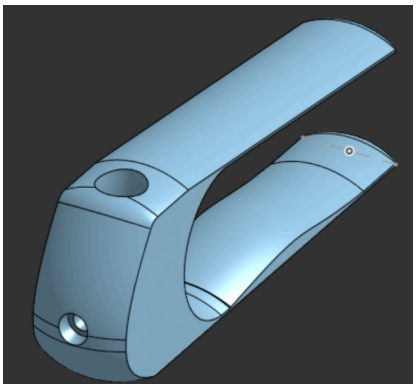
Improved design #1 (Silas):



This design kept the shape of the hand in mind, while also adding an alternative grip for the user in case they disliked or had issues using the original grip. This secondary grip also serves as storage for the pen. It was designed for the standard G2 pen, but can also accommodate pencils in the main slot. The pen is secured in the main slot via a hex nut inserted into the highlighted area (right image). Then, a standard 10-32 is inserted to secure the pen. In the future, there will be a grip similar to the key holder that will allow the user to secure any pen without needing a screwdriver for the 10-32. The idea of a brass heat insert has also been entertained, in case the user has issues with losing the nut.

Improved design #2 (Mitchell):





My design was designed around a Pilot G2 pen, which is fastened by a 10-32 screw that is held in by a brass insert. I focused mainly on the ergonomics of the handle and the thickness of the handle and the roof above the handle. I'll have to make the handle bigger on the bottom with my next iteration and most likely edit the hand groove on the top of the handle.

#3 - Bread/Bread Bagging tool